Preparations for Full Coverage Crowns

10

There are numerous situations that call for the use of a full coverage restoration. Clinicians have long considered it to be the most retentive of crown preparations.[1] Controlled laboratory studies have shown that when compared with partial coverage designs, the full coverage crown exhibits superior retention[2,3] and resistance.[4] This does not mean that a full coverage design must be used on every tooth. Instead, it should be used on those teeth whose restoration demands maximum retention.

A requirement for maximum retention is not commonly seen in the placement of single restorations. This need is more likely to be manifested in the design of retainers for fixed partial dentures, where additional demands are placed on the preparation and restoration. The selection of a full coverage crown retainer becomes mandatory when the abutment tooth is small or when the edentulous span is long.

Variations of the full coverage crown, the metal-ceramic crown and the all-ceramic crown, are used in situations that require a good esthetic result. The full coverage crown should be used when less extensive and less destructive designs have been considered and found lacking in retention, resistance, coverage, or esthetics to properly restore the tooth.

Full coverage in the right circumstances can be excellent treatment, but it is overused. More conservative dentistry has become easier, and the preservation of tooth structure is important.[5] A report of dental insurance data in 1979 indicated that nearly 93% of the cast restorations done by dentists submitting claims to that company were of a full-coverage design.[6] Undoubtedly, the percentage would be greater today. The indications for full coverage have increased because of the preservation of teeth that may have been removed in the past. Further, as patients' life expectancies have increased, so has the cumulative damage to teeth. There is also a greater need for the protection afforded by encircling the vulnerable parts of the endodontically treated tooth with full coverage.[7] However, the removal of all morphologic form of the tooth is radical treatment, and restoring it properly can be difficult.[8] The dentist should be certain it is necessary before placing a full coverage restoration.

Full Metal Crown Preparation

When all of the axial surfaces of a posterior tooth have been attacked by decalcification or caries or have been previously restored, the tooth is a candidate for a full metal crown. By tying together the remaining tooth structure, a full metal crown can strengthen and support the tooth.[9] It should be used judiciously, however, because it does require a destructive preparation. It may weaken rather than strengthen the remaining tooth structure when there has been previous extensive destruction in the center of the tooth. However, the preparation for a full metal crown is less destructive than that required for either a metal-ceramic or an all-ceramic crown.

Full coverage should not be used in mouths with uncontrolled caries. A full coverage crown is a restoration that replaces lost tooth structure and imparts some measure of structural support to the tooth. However, it does not protect the tooth against the biologic causes of the caries. These processes must be controlled by other means before any restoration can be successful.

Armamentarium

- Handpiece
- No. 171L bur
- Coarse-grit round-end tapered diamond (6856-016)
- Fine-grit round-end tapered diamond (8856-016)
- Medium-grit short needle diamond (852-012)
- Coarse-grit tapered torpedo diamond (6877K-014)
- Fine-grit tapered torpedo diamond (8877K-014)
- Red utility wax

Technique

The preparation for a full coverage crown is begun with occlusal reduction, creating about 1.5 mm of clearance on the functional and nonfunctional cusps. By accomplishing this step first, the occlusogingival length of the preparation can be determined. The potential retention of the preparation can then be assessed, and auxiliary features can be added if necessary.

Depth-orientation grooves are placed on the occlusal surface of the tooth to provide an easy reference to determine when reduction is sufficient. If reduction is begun without orientation marks, time will be wasted in repeated checks for adequate clearance. A coarse-grit round-end tapered diamond is used to place the grooves on the ridges and the primary grooves of the occlusal surface. If there is al-

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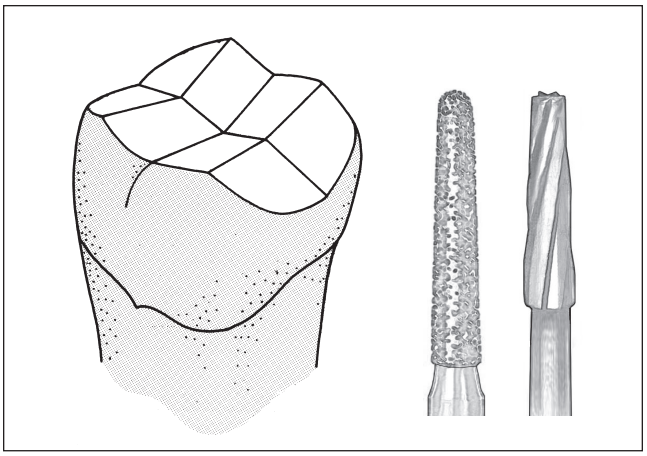


Fig 10-1 Occlusal reduction: coarse-grit round-end tapered diamond and no. 171L bur.

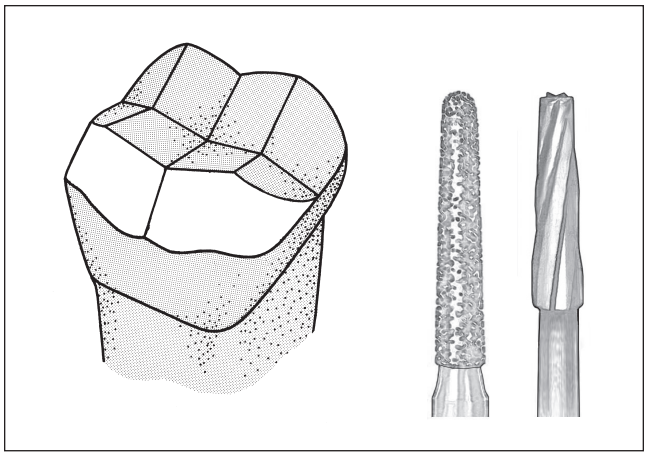


Fig 10-2 Functional cusp bevel: coarse-grit round-end tapered diamond and no. 171L bur.

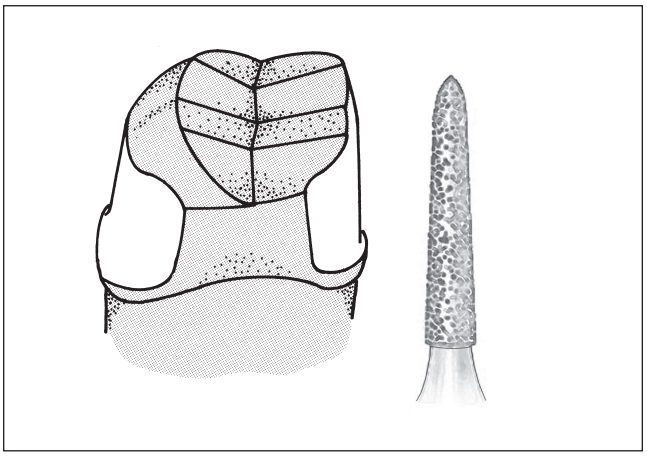


Fig 10-3 Facial and lingual axial reduction: coarse-grit tapered torpedo diamond.

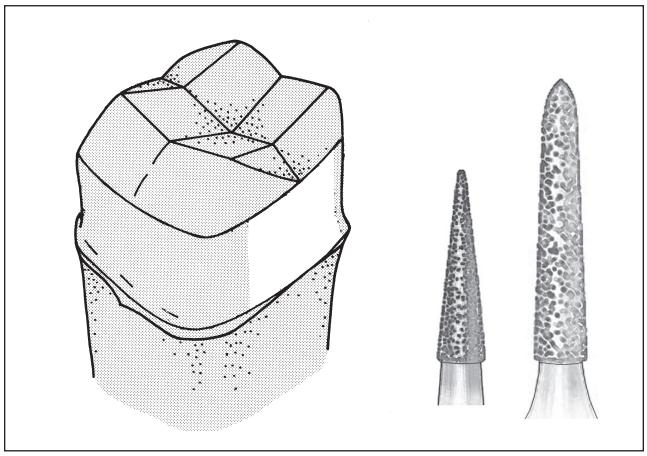


Fig 10-4 Proximal axial reduction: medium-grit short needle and coarse-grit tapered torpedo diamonds.

ready some clearance with the opposing tooth because of malpositioning or fracture of the tooth being prepared, the grooves need not be as deep.

The tooth structure remaining between the orientation grooves is removed to accomplish the occlusal reduction (Fig 10-1). Any roughness left by the grooves should be removed, keeping the occlusal surface in the configuration of the geometric inclines that make up the occlusal surface of any posterior tooth.

A wide bevel is placed on the functional cusp, again using the coarse-grit round-end tapered diamond (Fig 10-2). Depth-orientation grooves are also helpful in obtaining this reduction. The functional cusp bevel, placed on the facial inclines of mandibular facial cusps and the palatal inclines of maxillary palatal cusps, is an integral part of the occlusal reduction. Failure to place this bevel can produce a thin casting or poor morphology in the restoration.

Occlusal clearance is checked by having the patient close on a 2-mm-thick strip of red utility wax held over the preparation. The wax is then held up to a light to determine the adequacy of the occlusal clearance. Any part of the preparation that has insufficient occlusal clearance will be readily detectable as a thin spot in the wax. Additional tooth structure should be removed from the indicated areas and rechecked.

The occlusal reduction and functional cusp bevel are planed smooth with a no. 171L bur now or when the bur is used to instrument the seating groove. This can also be done with a fine-grit round-end tapered diamond. There should not be sharp angles or ridges where the planes or bevel join. If there are, they should be removed.

The facial and lingual walls are reduced with a coarse-grit tapered torpedo diamond, whose sides produce the desired axial reduction, while its tapered tip forms a chamfer finish line (Fig 10-3). A definite, even finish line is necessary to enable the fabrication of a restoration with a good fit,[10] and the chamfer is the best for providing the bulk needed for strength while still allowing good adaptation.[1]

The initial proximal cuts are made with a short needle diamond (Fig 10-4). The thin diamond is worked through the proximal area in an occlusogingival or faciolingual sawing

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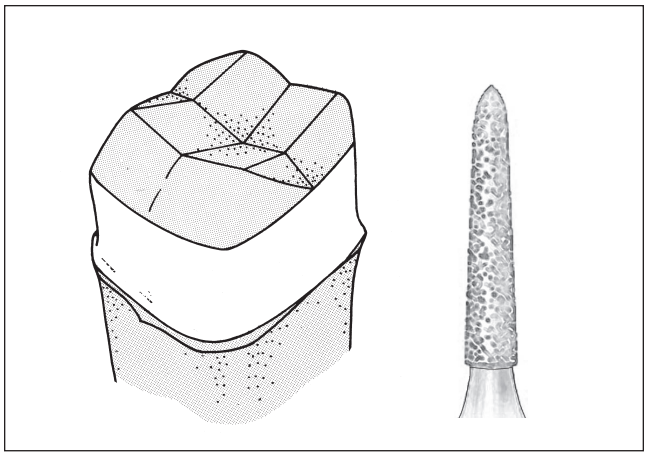


Fig 10-5 Chamfer and axial finishing: fine-grit tapered torpedo diamond.

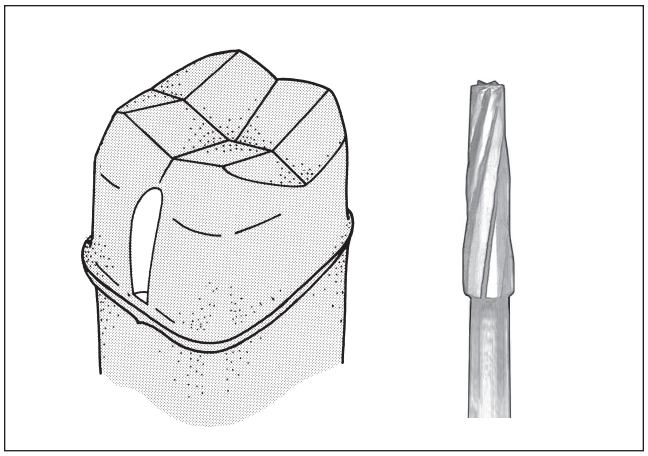


Fig 10-6 Seating groove: no. 171L bur.

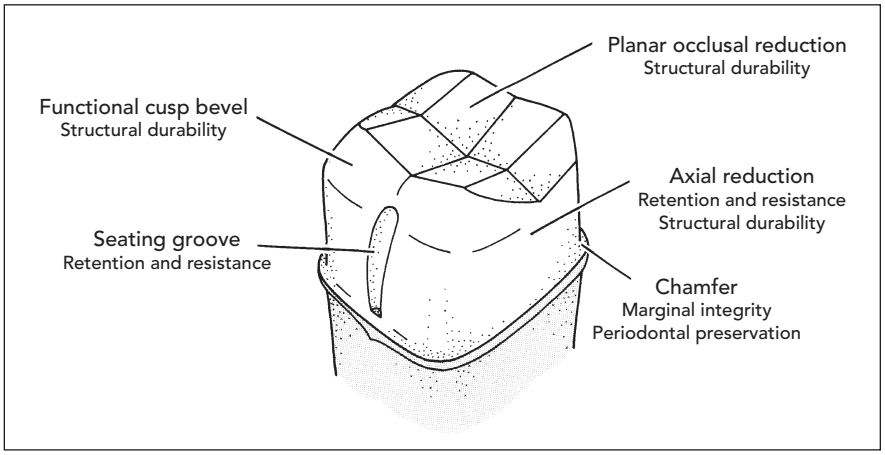


Fig 10-7 Features of a mandibular full metal crown preparation and the function(s) served by each.

motion, carefully avoiding the adjacent teeth. Once sufficient maneuvering room has been obtained, the coarse-grit tapered torpedo diamond is introduced to plane the walls while simultaneously forming a chamfer as the interproximal gingival finish line.

All of the axial surfaces are smoothed with a fine-grit tapered torpedo diamond, whose size and shape enable it to finish the chamfer finish line as well (Fig 10-5). Special care should be taken in rounding the corners from the facial or lingual surfaces to the proximal surfaces to ensure that the finish line will be smooth and continuous.

The final step in the full coverage crown preparation is the placement of a seating groove (Fig 10-6). It will prevent any rotational tendencies during cementation, and it will help guide the casting into place. The groove is formed with a no. 171L bur and is placed in the axial surface with the greatest bulk. This usually will be on the facial surface of mandibular preparations and on the palatal surface of maxillary preparations. On preparations for long-span fixed partial dentures, there should be both a facial and a lingual groove to increase the resistance to mesiodistal movement. The features of a preparation for a full veneer metal crown and the function served by each are shown in Fig 10-7.

Metal-Ceramic Crowns

The metal-ceramic restoration, also called a *porcelain-fused-to-metal restoration*, consists of a ceramic layer bonded to a thin cast metal coping that fits over the tooth preparation. Such a restoration combines the strength and accurate fit of a cast metal crown with the cosmetic effect of a ceramic crown. With a metal understructure, metal-ceramic restorations have greater strength than restorations made of ceramic alone. Friedlander et al[11] found the metal-ceramic

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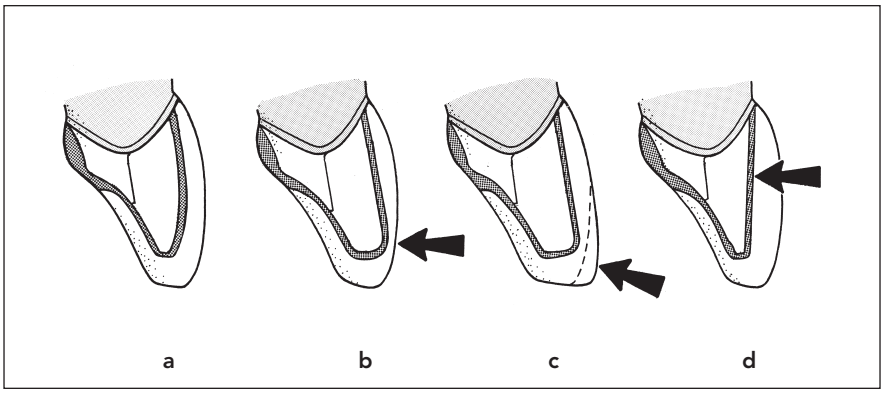


Fig 10-8 It is important to reduce the labial surface in two planes to receive a metal-ceramic restoration *(a)*. If only one plane is reduced, opaque porcelain may show through *(b)*, the labial surface may be overcontoured *(c)*, or the pulp may be encroached upon *(d)*.

restoration to be 2.8 times as strong as ceramic restorations. As a result, the longevity of metal-ceramic restorations is greater and it can be used in a wider variety of situations, including the replacement of missing teeth with fixed partial dentures.

Because the restoration is a combination of metal and ceramic, it is not surprising that the tooth preparation for it is likewise a combination. There is deep reduction on the facial surface to provide space for the coping and a ceramic layer thick enough to achieve the desired esthetic result. On the lingual surface, fading proximally, there is shallower reduction similar to that used for a full metal crown. There may be a wing on each proximal surface where the deep facial reduction ends and the shallower proximal reduction begins.[12] The transition between these reductions need not be dramatic, though, because extreme concentration on the necessity of a wing often results in inadequate axial reduction for metal or excessive axial reduction for ceramic. Rather than being sharply delineated, the transition is often gradual. This is especially true for anterior teeth because of their narrow interproximal contact areas and diminished faciolingual dimension.

Adequate reduction is essential to achieving a good esthetic result. Without the space for a sufficient thickness of ceramic material, two things can happen: *(1)* The restoration will be poorly contoured, adversely affecting both the esthetic effect of the crown and the health of the surrounding gingiva, and/or *(2)* the shade and translucency of the restoration will not match adjacent natural teeth.

Anterior metal-ceramic crowns

A uniform reduction of approximately 1.2 mm is needed over the entire facial surface. To achieve adequate reduction without encroaching on the pulp, the facial surface must be prepared in two planes that correspond roughly to the two geometric planes present on the facial surface of an uncut tooth (Fig 10-8). If the facial surface is reduced in one plane that is an extension of the gingival plane, the incisal edge will protrude, resulting in a bad shade match or an overcontoured block. If the facial surface is prepared in one plane that has adequate facial reduction in the incisal aspect, the facial surface will be overtapered and too close to the pulp.

Armamentarium

- Laboratory knife with no. 25 blade
- Silicone putty and accelerator
- Handpiece
- Coarse-grit flat-end tapered diamond (6847-016)
- Coarse-grit football-shaped diamond (6379-023)
- Medium-grit long needle diamond (850-012)
- Coarse-grit tapered torpedo diamond (6877K-014)
- Fine-grit tapered torpedo diamond (8877K-014)
- Fine-grit flat-end tapered diamond (8847KR-016)
- CP-11/12 (Hu-Friedy)

Technique

An index made before the preparation is begun makes it possible to have a positive check on reduction produced by the preparation. If the contours of the existing tooth are correct, the index can be made intraorally while waiting for the anesthetic to take effect. However, if the tooth is badly broken down, or if contours are to be changed in the finished restoration, the index should be made from a preoperative wax-up on the diagnostic cast.

A half-scoop of putty is mixed with the appropriate amount of accelerator and kneaded in the palm of the hand until all streaks of the accelerator have disappeared. The putty is then adapted with a thumb and forefinger over the tooth to be prepared (Fig 10-9). It should be allowed to polymerize on the tooth, which should take about 2 minutes. The index should cover the entire labial and lingual surface of the tooth to be prepared, plus the corresponding surfaces of at least one adjacent tooth (Fig 10-10).

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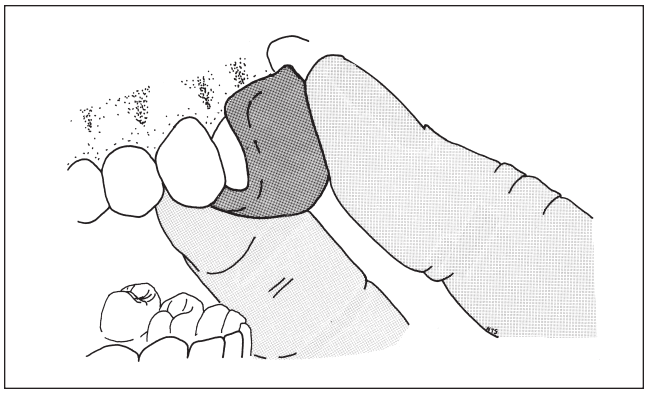


Fig 10-9 Silicone putty is molded on the labial and lingual surfaces of the tooth to be prepared and the teeth adjacent to it.

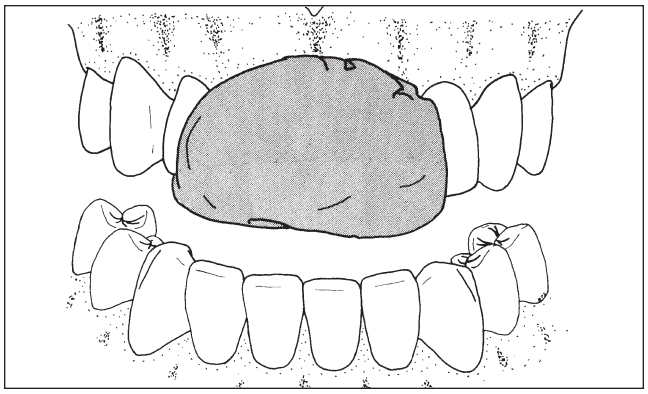


Fig 10-10 The index should contact the labial and lingual surfaces of the teeth on either side of the tooth to be prepared.

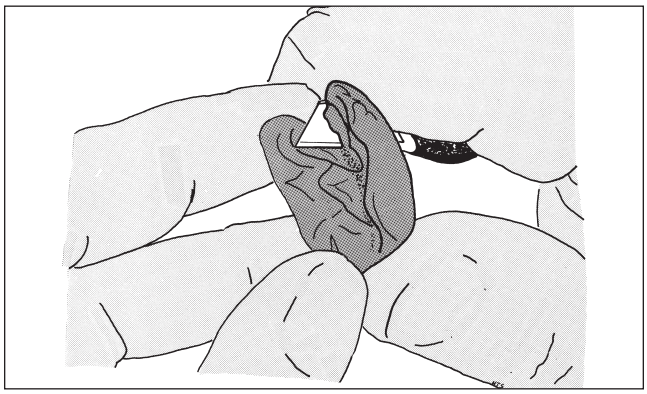


Fig 10-11 The index is cut into a labial and a lingual half.

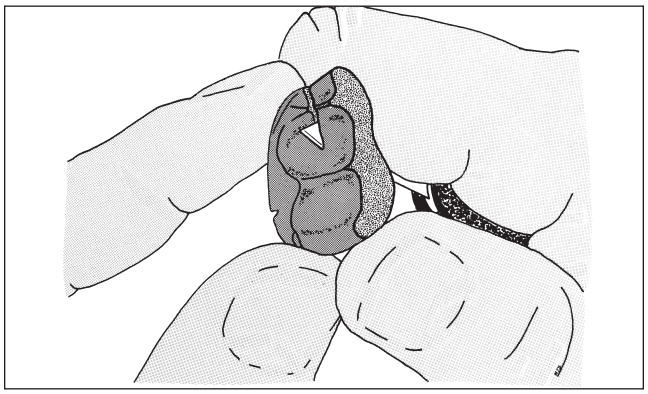


Fig 10-12 The labial half of the index is cut into a gingival and an incisal half.

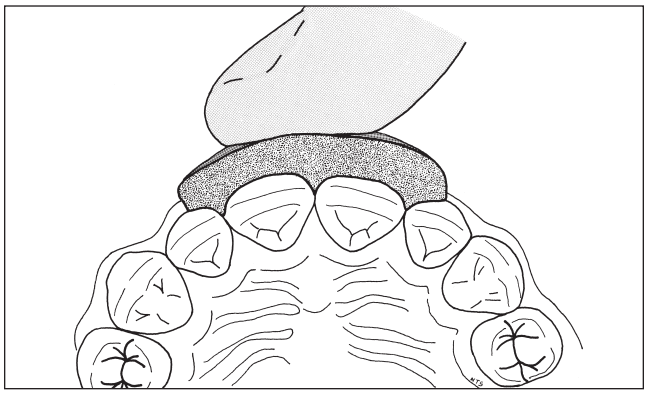


Fig 10-13 The gingival half of the index is placed on the teeth to see if it fits accurately.

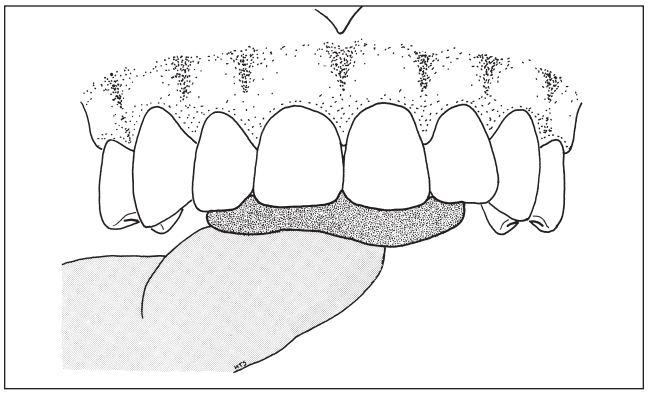


Fig 10-14 The lingual index is placed in position to check its accuracy.

The index is then removed from the teeth. A laboratory knife with a no. 25 blade is used to cut along the incisal edges of the tooth imprints to separate the index into a labial and a lingual half (Fig 10-11). The lingual half is set aside for the time being. The labial portion of the index is cut from mesial to distal across the imprints of the labial surfaces of the teeth to produce an incisal half and a gingival half (Fig 10-12).

The gingival half of the labial portion is positioned on the teeth to ensure that it is closely adapted to the labial surfaces (Fig 10-13). After removing the labial index, the lingual index is put in position, and its adaptation to the incisal edges of the teeth is checked (Fig 10-14).

The labial and lingual indices are set aside until the preparation is completed. Then the gingival half of the labial index

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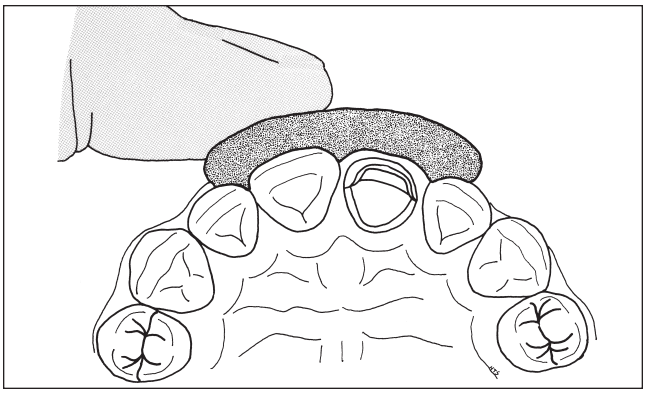


Fig 10-15 The gingival half of the index is positioned over the preparation to check the labial reduction.

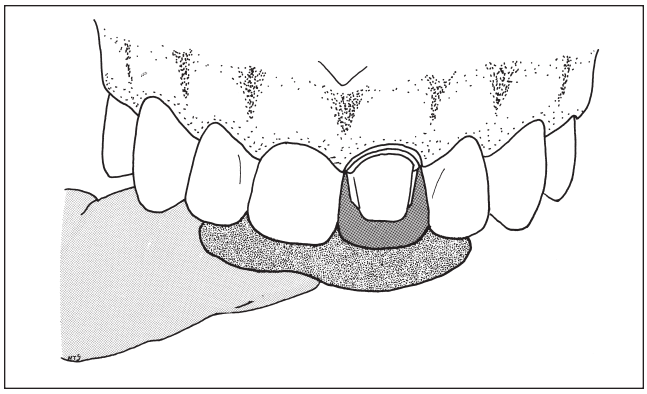


Fig 10-16 The lingual index is placed in position to evaluate the incisal reduction.

is positioned and checked for adequate labial clearance for a metal coping and porcelain (Fig 10-15). If the reduction is inadequate, the index is removed from the mouth and more tooth structure is removed. The incisal clearance is checked by putting the lingual index in place and evaluating the distance between the incisal edge of the prepared tooth and the incisal edge of the tooth imprint on the index (Fig 10-16).

The initial step in the preparation for a metal-ceramic crown is the placement of depth-orientation grooves on the labial and incisal surfaces with a coarse-grit flat-end tapered diamond. These orientation cuts, recommended by Preston[12] and Miller,[13] are a means of judging the amount of tooth structure to be removed. The full diameter of an instrument of known dimension is sunk into the tooth with light pressure, keeping the diamond moving incisogingivally. Putting excessive pressure on the handpiece and burying the diamond in the facial surface causes excessive heat that will likely result in pulpal damage. Lockard[14] attributes his lack of pulpal sequelae to using new diamonds with very light pressure and a constant sweeping motion.

The depth of reduction can be measured using the uncut outer surface of the remaining tooth structure as a reference point.[13] If reduction is done without grooves, time will be wasted in repeatedly rechecking reduction with the index. The labial grooves should be cut in two sets: one set parallel with the gingival half of the labial surface and one set parallel with the incisal half of the labial surface (Fig 10-17). These grooves should be 1.2 mm deep. The incisal grooves should be cut all the way through the incisal edge and should extend 2.0 mm gingivally.

Incisal reduction is done with the coarse-grit flat-end tapered diamond so that it parallels the inclination of the unprepared incisal edge (Fig 10-18). This is done first to allow easy instrument access to the axial surfaces and the gingival finish line. Inadequate incisal reduction results in poor incisal translucency in the finished restoration.

Reduction of the incisal portion of the labial surface is done with the same flat-end tapered diamond. All tooth structure is planed off to the depth of the orientation grooves (Fig 10-19). The gingival portion of the labial surface is likewise reduced to the depth of the grooves with the flat-end tapered diamond. The reduction is carried around the labioproximal line angles to a point 1.0 mm lingual to the proximal contacts (Fig 10-20). Although the resulting wings of tooth structure can provide some resistance to rotation, that is not the primary reason for their existence. They conserve tooth structure, if in fact there is still sound tooth structure left on the proximal surfaces. It is important that the portion of each wing that faces labially has the same inclination as the gingival portion of the labial surface; otherwise, an undercut will be created.

The lingual surface (incisal to the cingulum) is reduced with a coarse-grit football-shaped diamond to obtain a minimum of 0.7 mm of clearance with the opposing teeth (Fig 10-21). Those portions of the lingual surface that will have a ceramic veneer should have 1.0 mm of clearance. The junction between the cingulum and the lingual wall (apical to the cingulum) must not be overreduced. Overshortening the lingual wall will reduce retention.

A long needle diamond is used to complete access through the proximal areas to minimize the chances of nicking the adjacent teeth (Fig 10-22). Much of the axial reduction in the region of the proximal contact will have been accomplished already by the flat-end tapered diamond. The lingual aspect of the proximal axial walls, as well as the lingual surface, are reduced with the coarse-grit tapered torpedo diamond (Fig 10-23). The lingual and proximal axial surfaces are smoothed with the fine-grit tapered torpedo diamond, accentuating the chamfer on the lingual and proximal surfaces at the same time (Fig 10-24).

A fine-grit flat-end tapered diamond bur is used to smooth the labial surface (Fig 10-25). All angles and edges on the preparation are rounded with the sides of the diamond to facilitate seating of the restoration later. At the same time that the labial surface is being planed by the side of the diamond, the end is forming a radial shoulder finish line.

Shoulders have been advocated for gingivofacial finish lines of metal-ceramic preparations alone[15–19] or with narrow

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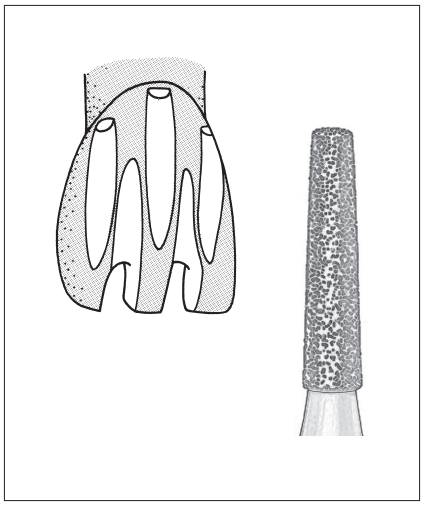


Fig 10-17 Depth-orientation grooves: coarse-grit flat-end tapered diamond.

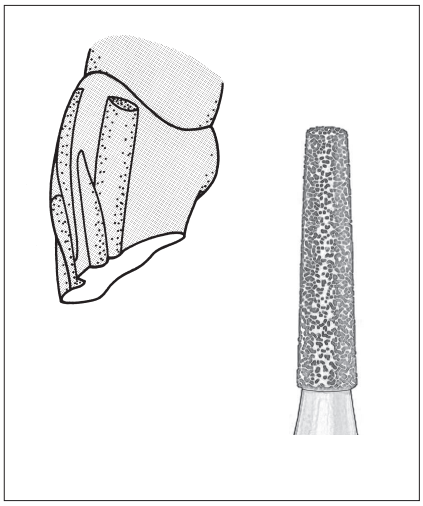


Fig 10-18 Incisal reduction: coarse-grit flat-end tapered diamond.

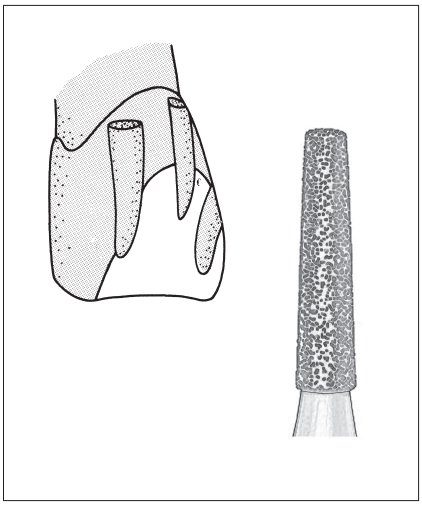


Fig 10-19 Labial reduction (incisal half): coarse-grit flat-end tapered diamond.

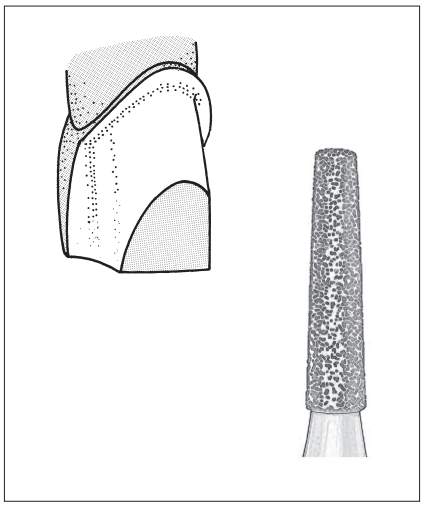


Fig 10-20 Labial reduction (gingival half): coarse-grit flat-end tapered diamond.

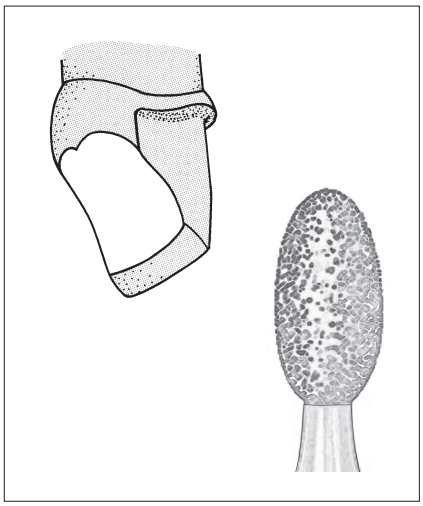


Fig 10-21 Lingual reduction: coarse-grit football-shaped diamond.



Fig 10-22 Initial proximal reduction: medium-grit long needle diamond.

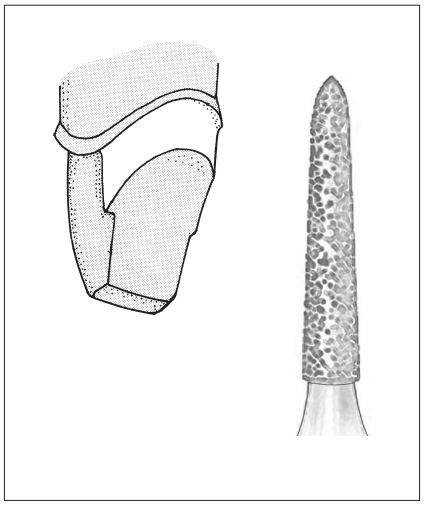


Fig 10-23 Lingual axial reduction: coarse-grit tapered torpedo diamond.

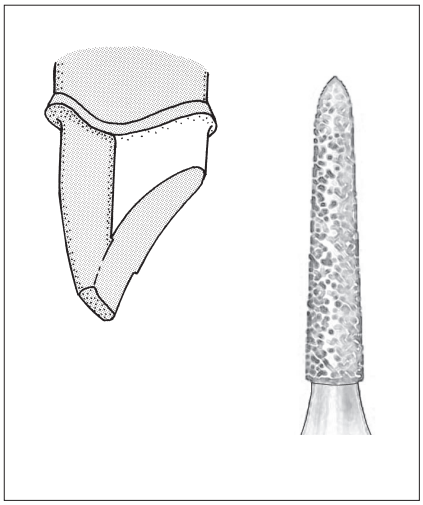


Fig 10-24 Axial finishing: fine-grit tapered torpedo diamond.

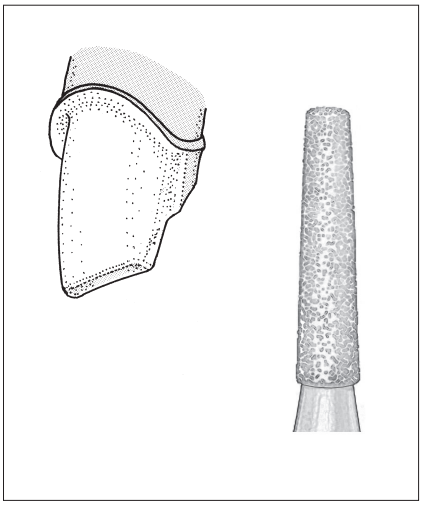


Fig 10-25 Axial and shoulder finishing: fine-grit flat-end tapered diamond.

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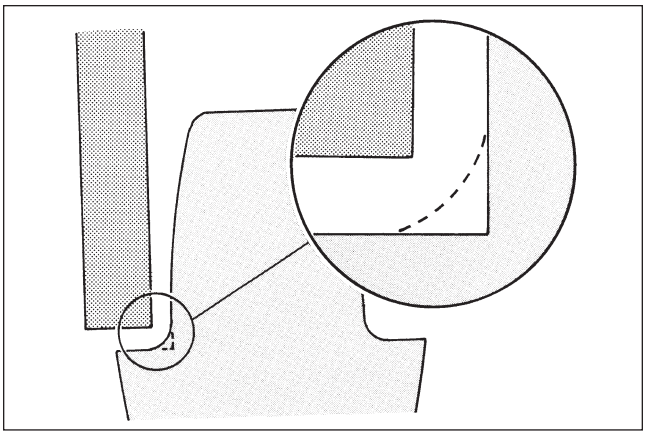


Fig 10-26 The sharp corners of a conventional chisel will gouge the gingivoaxial angle (inset) of a radial shoulder.

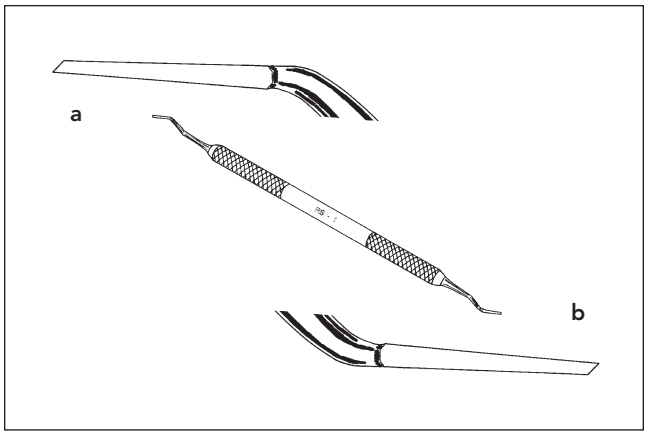


Fig 10-27 *(a and b)* The CP-11/12 is a modified 15-8-8 chisel with the same hoe (pull stroke) blade on both ends.

bevels.[12,20–23] Some investigators have reported that metal-ceramic crowns with metal gingivofacial margins made over shoulder finish lines distort less during porcelain firing.[24,25] A possible explanation is that the shoulder configuration provides space for an internal rib of metal to buttress the margin.[12,26] Other investigators have reported not finding a difference in marginal fit. They hypothesize that marginal gaps following ceramic firing may be caused either by technical difficulties in forming a knife edge of metal and ceramic[27] or by differences in metal-ceramic combinations.[28]

However, there is a compelling reason for not using a metal margin at all. The metal collar that accompanies a bevel on a shoulder[29] often requires the finish line to be placed deep in the gingival sulcus to hide the metal.[30] If some form of shoulder without a bevel is used, an all-ceramic margin can be fabricated. This eliminates a metal collar at the faciogingival margin of the finished metal-ceramic restoration, and there is no need to bury the margin beneath the gingiva. Quantitative evaluations of the marginal fit of all-ceramic shoulders on metal-ceramic crowns have found satisfactory adaptation of the ceramic to the preparation finish line. Belser et al[31] reported in vivo marginal discrepancies of 46 µm on cemented metal-ceramic crowns with all-ceramic facial margins. They found no significant differences among crowns with all-ceramic margins and those with metal collar margins over shoulder and beveled shoulder finish lines. In vitro studies by West and associates[32] and Hunt et al[33] found minimal marginal discrepancies in all-ceramic shoulder margins on metal-ceramic crowns. Of course, a dentist can use all-ceramic margins on metal-ceramic crowns only if the technician is capable of producing restorations with accurate ceramic margins.

Zena et al[34] demonstrated that all-ceramic margins made over hand-planed shoulders fit significantly better than margins made over finish lines cut solely with rotary instruments. However, if a conventional enamel chisel is used for planing a radial shoulder, the sharp angles at the ends of the cutting blade will destroy the rounded internal angle of the finish line (Fig 10-26).

A modified 15-8-8 binangle chisel, the CP-11/12, is recommended to avoid this problem (Fig 10-27). This instrument has a hoe (pull stroke) blade at both ends, unlike a conventional 15-8-8 binangle chisel, which has a hoe (pull stroke) blade at one end and a chisel (push stroke) blade at the other (Fig 10-28). One corner of a CP-11/12 blade is rounded with a mounted Arkansas stone (Fig 10-29a), and the opposite corner is rounded on the other end (Fig 10-29b).

One end, with the rounded corner against the gingivoaxial junction, is used to instrument the radial shoulder on half of the preparation (Fig 10-30). The other end, with its modified corner also against the rounded gingivoaxial junction, is used to smooth the finish line on the other half of the preparation. The 1.5-mm-wide blade will extend over the actual finish line, which is 1.2 to 1.5 mm wide. This will remove any lip of enamel that might extend incisally from the cavosurface angle.

The features of a preparation for an anterior metal-ceramic restoration and the function served by each are shown in Fig 10-31.

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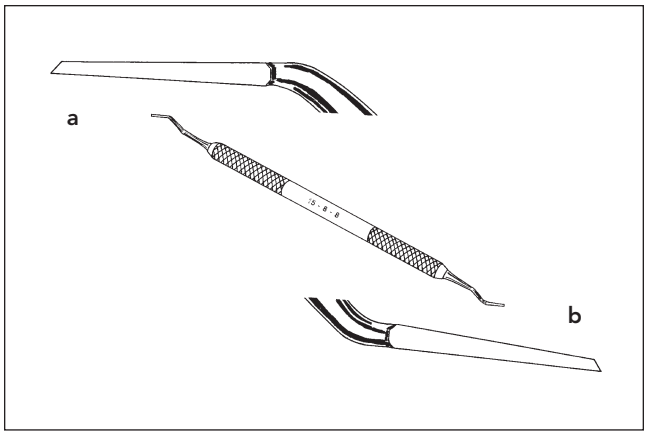


Fig 10-28 A conventional 15-8-8 binangle chisel has a hoe (pull stroke) blade on one end *(a)* and a chisel (push stroke) blade on the other *(b)*.

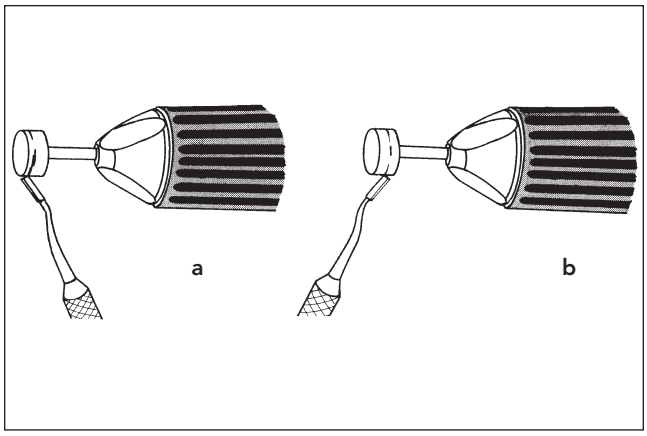


Fig 10-29 Round an angle at one end of the CP-11/12 *(a)* and the opposite angle on the other end *(b)*.

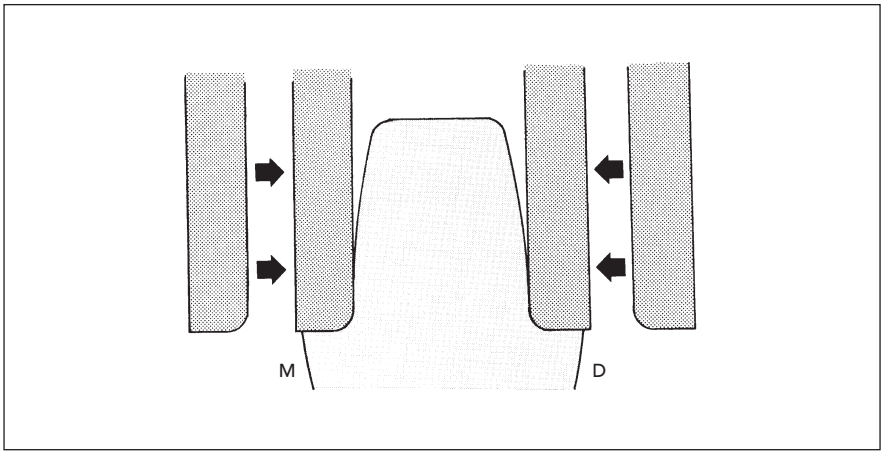


Fig 10-30 The rounded corner of one end of the CP-11/12 is placed against the gingivoaxial angle while planing the mesial half of the shoulder (M). The other end is used to instrument the distal half of the finish line (D).

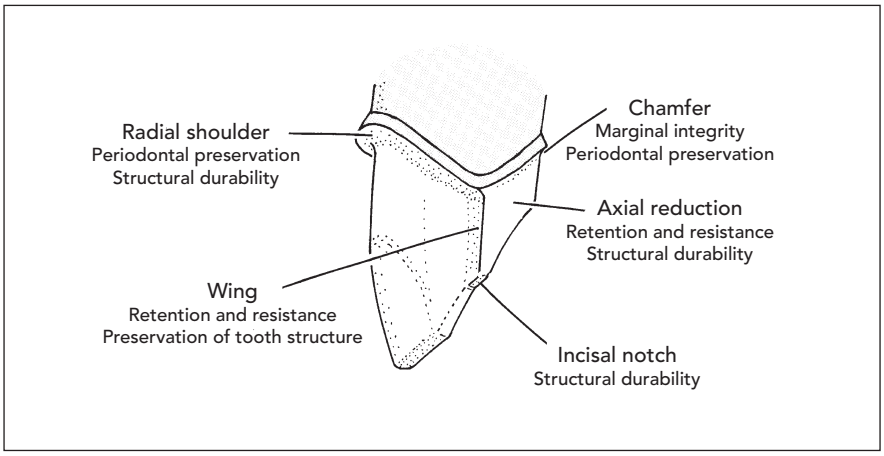


Fig 10-31 Features of an anterior metal-ceramic preparation and the function(s) served by each.

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Posterior metal-ceramic crowns

The use of metal-ceramic crowns on posterior teeth allows the creation of an esthetic restoration on a posterior tooth needing a full crown. Maxillary premolars and first molars and mandibular first premolars are almost always visible in the full smile. Mandibular second premolars also can fall into this category. Maxillary second molars and mandibular molars may require metal-ceramic crowns if a patient will not accept all-metal crowns on those teeth, although they are rarely seen in most mouths.

Routinely placing metal-ceramic crowns on all premolars and molars is overtreatment because of the additional tooth structure that must be destroyed to accommodate the combined thickness of metal and ceramic. Often, there is added expense for the patient because of higher laboratory fees as well as an increased risk of failure from ceramic veneer fracture.

The routine use of all-ceramic occlusal surfaces has been criticized.[35] This restoration design offers maximum esthetic effect when required by location in a highly visible area or by patient preference. Patients who demand ceramic occlusal surfaces should know of the potential problems. The use of all-ceramic occlusal surfaces requires the removal of more tooth structure, and the completed restorations pose a threat to the structural integrity of opposing occlusal surfaces. Conventional glazed dental porcelain is approximately 40 times as abrasive as gold to tooth enamel.[36] Some newer, ultra-low-fusing ceramics have shown improved wear rates against enamel.[37] However, their abrasiveness is still significantly higher than that of polished metal.[37] Preparations for metal-ceramic crowns should be done with a plan for the extent of ceramic coverage in mind because the areas to be veneered with ceramic require deeper reduction than those portions of the tooth that will be overlaid with metal alone.

Armamentarium

- Laboratory knife with no. 25 blade
- Silicone putty and accelerator
- Handpiece
- Coarse-grit round-end tapered diamond (6856-016)
- Fine-grit round-end tapered diamond (8856-016)
- Coarse-grit flat-end tapered diamond (6847-016)
- Medium-grit short needle diamond (852-012)
- Coarse-grit tapered torpedo diamond (6877K-014)
- Fine-grit tapered torpedo diamond (8877K-014)
- Fine-grit flat-end tapered diamond (8847KR-016)
- Coarse-grit flame diamond (862-010)
- Fine-grit flame diamond (8862-010)
- CP-11/12 binangle chisel

Technique

Before the preparation is begun, silicone putty is adapted to the facial, lingual, and occlusal surfaces of the tooth to be prepared as well as to one tooth on each side. After polymerization, a midsagittal index can be formed by cutting the silicone in half along the faciolingual midline of the tooth to be prepared. The putty is placed back on the tooth to ensure good adaptation. If the clinical crown of the tooth being restored is severely damaged, the index should be made from a diagnostic wax-up.

A facial index is made by cutting through the silicone along the facial cusps of the teeth. The facial piece is divided along a line midway between the cervical lines of the teeth and the facial cusp tips. The occlusal portion is discarded, and the gingival portion is used as an index.

The occlusal reduction is begun by making depth-orientation grooves with a round-end tapered diamond. In the areas where there will be ceramic coverage, reduction should ultimately be 2.0 mm.[12,20,38,39] The occlusal reduction is completed by removing the strips of intact enamel between the depth-orientation grooves with the same diamond. The reduction should take the form of definite planes reproducing the general occlusal morphology[40] or the basic geometric shape of the occlusal surface (Fig 10-32).

The functional cusp bevel, which allows a uniform bulk of restorative material on the palatal inclines of maxillary palatal cusps and the facial inclines of mandibular facial cusps, is also begun with depth-orientation grooves (Fig 10-33). The depth required is 1.5 mm if the coverage will be metal only and 2.0 mm if the metal will be veneered with ceramic. The functional cusp bevel is completed by removing the tooth structure between the depth-orientation grooves. The angulation of the bevel approximates the inclination of the opposing cusps.

A no. 171L bur or fine-grit round-end tapered diamond is used to smooth the planes of the occlusal reduction to remove any roughness or pits that might interfere with the complete seating of the finished restoration. Any sharp corners or edges on the preparation that might cause problems in impression pouring, investing, casting, and ultimately in the seating of the completed crown should be rounded over.

The coarse-grit flat-end tapered diamond is aligned with the occlusal segment of the facial surface, and three vertical grooves are cut in the occlusal portion of the facial surface. These are nearly the full diameter of the instrument, fading out gingivally (Fig 10-34). The same diamond is aligned with the gingival component of the facial surface, and the side of the instrument is used to cut into the tooth surface. The full diameter of the instrument must cut into the tooth. The instrument tip should be slightly supragingival at this point, even if the intended location of the finish line is flush with or slightly below the gingival crest. At least two more orientation grooves should be placed near the line angles of the tooth.

All tooth structure remaining between the depth-orientation grooves in the occlusal segment of the facial surface is removed with the flat-end tapered diamond (Fig 10-35). The gingival portion of the facial surface is then reduced, extending it well into the proximal surface (Fig 10-36). If facial reduction of less than 1.2 mm is done for a base metal–ceramic crown or 1.4 mm for a noble metal–ceramic crown, the restoration will be either opaque or overcontoured.

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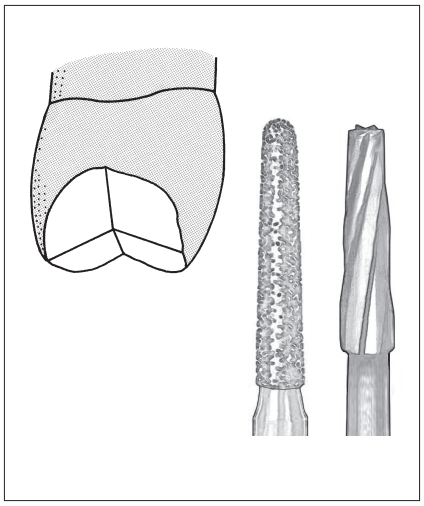


Fig 10-32 Planar occlusal reduction: coarse-grit round-end tapered diamond and no. 171L bur.

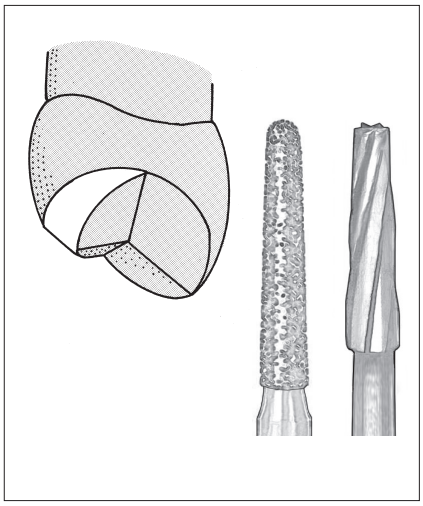


Fig 10-33 Functional cusp bevel: coarse-grit round-end tapered diamond and no. 171 bur.



Fig 10-34 Depth-orientation grooves: coarse-grit flat-end tapered diamond.

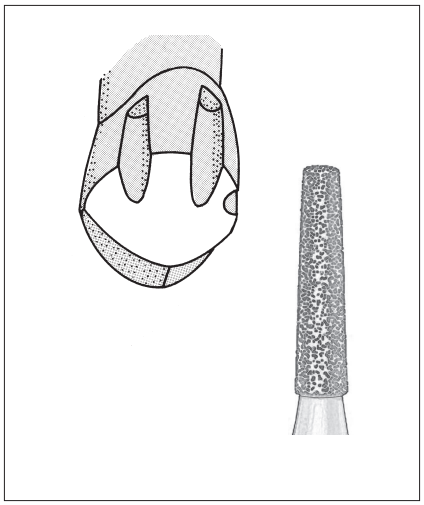


Fig 10-35 Facial reduction (occlusal half): coarse-grit flat-end tapered diamond.



Fig 10-36 Facial reduction, (gingival half): coarse-grit flat-end tapered diamond.

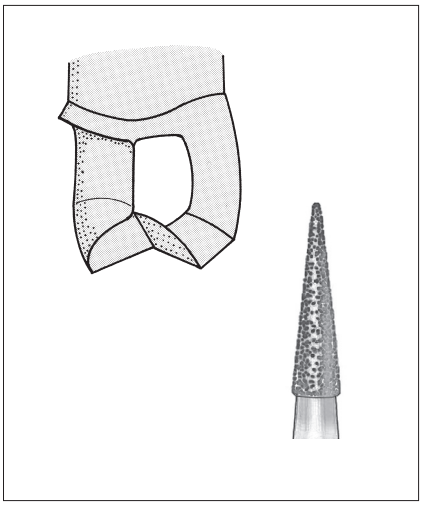


Fig 10-37 Proximal axial reduction: medium-grit short needle diamond.

The proximal axial reduction is begun with a short needle diamond (Fig 10-37). Its narrow diameter allows interproximal reduction without nicking adjacent teeth. The instrument can be used with an up-and-down motion on the facial aspect of the interproximal tooth structure, or it can be used on the occlusal portion with a faciolingual movement. Initially, the objective is to achieve separation between the teeth without overtapering the prepared walls or mutilating the adjacent tooth. The proximal axial surfaces are then planed with the needle diamond.

The lingual axial wall is reduced with a coarse-grit tapered torpedo diamond (Fig 10-38). Enough tooth structure is removed on both the lingual and proximal axial walls to create a distinct chamfer finish line wherever there will not be a ceramic veneer. The chamfer finish line and the axial surfaces adjacent to it are smoothed with a fine-grit tapered torpedo diamond. All axial surfaces that will be veneered only with metal are finished in this way.

The facial surface and those parts of the proximal surfaces to be veneered with ceramic are smoothed with a fine-grit

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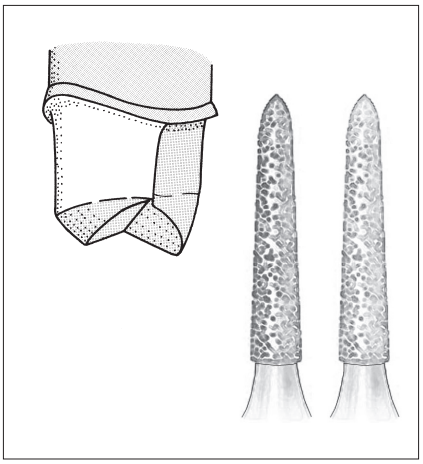


Fig 10-38 Lingual axial reduction and finishing: coarse- and fine-grit tapered torpedo diamonds.

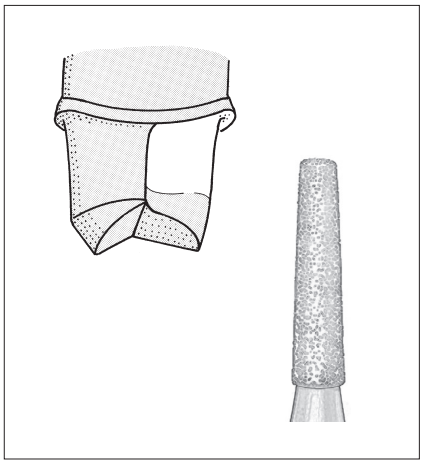


Fig 10-39 Facial axial and radial shoulder finishing: fine-grit flat-end tapered diamond.

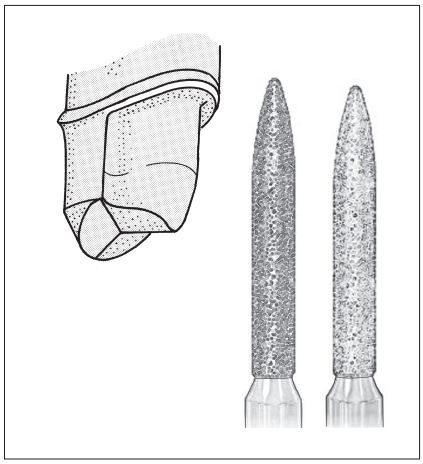


Fig 10-40 Gingival bevel: medium- and fine-grit flame diamonds.

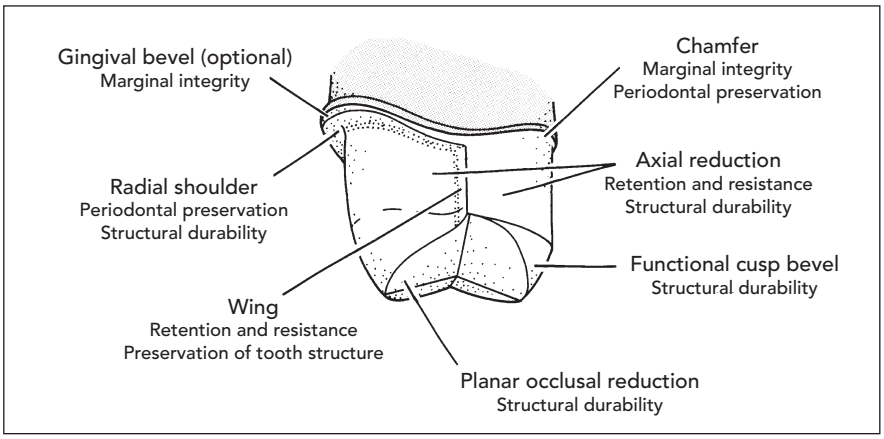


Fig 10-41 Features of a preparation for a metal-ceramic crown on a posterior tooth and the function(s) served by each.

flat-end tapered diamond (Fig 10-39). At the most lingual extension of the facial reduction, lingual to the proximal contact, the transition from the deeper facial reduction to the relatively shallower lingual axial reduction results in a vertical wall or "wing" of tooth structure. The wings must not be undercut with the facial or lingual axial walls of the preparation.

If the shoulder and wings are not lingual to the proximal contact, the proximal area of the ceramic veneer will lack translucence. If there was an amalgam restoration in the tooth prior to this preparation, the wing is made to coincide with the lingual wall of the amalgam's proximal box. If the entire proximal surface is to be veneered with ceramic, the shoulder is extended across the proximal surface with no wing.

The radial shoulder, started with the flat-end tapered diamond at the time the facial reduction was accomplished, is finished now with the radial shoulder fine-grit flat-end tapered diamond. On highly visible posterior teeth, such as the maxillary premolars, an all-ceramic margin is frequently used to achieve a good esthetic result without intruding into the gingival sulcus. The 1.0-mm-wide shoulder is smoothed by planing it with the CP-11/12 modified binangle chisel, which will preserve the rounded internal angle created by the radial shoulder diamond. Any "lip" or reverse bevel of enamel at the cavosurface angle should be removed. Small, sharp edges in this area may not be reproduced when the impression is poured, and they are susceptible to fracture on the cast or on the tooth in the mouth.

There are occasions when a shoulder with a bevel is the finish line of choice: when esthetic needs are not as critical or the dental technician is unable to consistently produce a precise all-ceramic margin. A narrow bevel, no wider than 0.3 mm, can be placed on the shoulder with the tip of a medium-grit flame diamond (Fig 10-40). The bevel should be kept narrow because the metal collar on the resulting crown must be as wide as the bevel. It is easier to wax and cast to the bevel if the diamond is leaned toward the center of the tooth as much as possible. The bevel is finished with a fine-grit flame diamond to create a finish line that is as clear as possible. The features of a preparation for a posterior metal-ceramic restoration and the function served by each are shown in Fig 10-41.

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Fig 10-42 Depth-orientation grooves: coarse-grit flat-end tapered diamond.

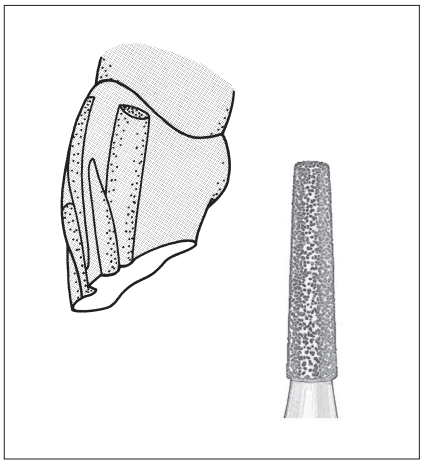


Fig 10-43 Incisal reduction: coarse-grit flat-end tapered diamond.

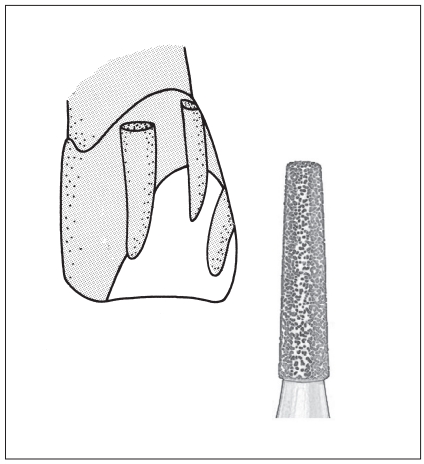


Fig 10-44 Labial reduction (incisal half): coarse-grit flat-end tapered diamond.

All-Ceramic Crowns

The all-ceramic crown differs from other cemented veneer restorations because it is not cast in gold or some other metal. It is capable of producing the best esthetic effect of all dental restorations. However, because it is made entirely of ceramic, a brittle substance, it is more susceptible to fracture. The development of dental porcelain reinforced with alumina in the 1960s created renewed interest in the restoration.[41] In recent years, this interest has mushroomed with the use of new reinforcement materials such as lithium disilicate and zirconia.[42,43]

Preparations for this type of crown should be left as long as possible to give maximum support to the porcelain. An overshortened preparation will create stress concentrations in the labiogingival area of the crown,[3] which can produce a characteristic half-moon fracture in the labiogingival area of the restoration.[44–46] A shoulder of uniform width (approximately 1 mm) is used as a gingival finish line to provide a flat seat to resist forces directed from the incisal.[44,47] The incisal edge is flat and placed at a slight inclination toward the linguogingival to meet forces on the incisal edge and prevent shearing.[48,49] Finally, all sharp angles of the preparation should be slightly rounded to reduce the danger of fracture caused by points of stress concentration.[44,48,49]

The position of the tooth in the arch, factors relating to occlusion, and morphologic features of the tooth all should be weighed when an all-ceramic crown is considered for a restoration. All-ceramic crowns are best suited for use on incisors. If they are used on other teeth, patients should know that there is an increased risk of fracture.

Use of the all-ceramic crown should be avoided on teeth with an edge-to-edge occlusion that will produce stress in the incisal area of the restoration. It likewise should not be used when the opposing teeth occlude on the cervical fifth of the lingual surface. Tension will be produced, and a half-moon fracture is likely to occur. Teeth with short cervical crowns also are high risks for all-ceramic crowns because they do not have enough preparation length to support the lingual and incisal surfaces of the restoration.

Armamentarium

- Handpiece
- Coarse-grit flat-end tapered diamond (6847-016)
- Coarse-grit football-shaped diamond (6379-023)
- Fine-grit flat-end tapered diamond (8847KR-016)
- CP-11/12 binangle chisel

Technique

Depth-orientation grooves are placed on the labial and incisal surfaces with the coarse-grit flat-end tapered diamond before any reduction is done (Fig 10-42). Without grooves it is impossible to accurately gauge the depth of reduction done on the labial surface. The grooves are 1.2 to 1.4 mm deep on the labial and 2.0 mm deep on the incisal. Three labial grooves are cut with the diamond held parallel to the gingival one-third of the labial surface. A second set of two grooves is made parallel to the incisal two-thirds of the uncut labial surface. The labial surface of an all-ceramic preparation is done in two planes to achieve adequate clearance for good esthetics without encroaching on the pulp.[48]

Incisal reduction is done with the coarse-grit flat-end tapered diamond so that it will be possible for instruments to reach the finish line area of the preparation in subsequent steps. Approximately 1.5 to 2.0 mm of tooth structure is removed (Fig 10-43).

The tooth structure still remaining between the depth-orientation grooves on the incisal portion of the labial surface is planed away (Fig 10-44). The gingival portion of the

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Fig 10-45 Labial reduction (gingival half): coarse-grit flat-end tapered diamond.

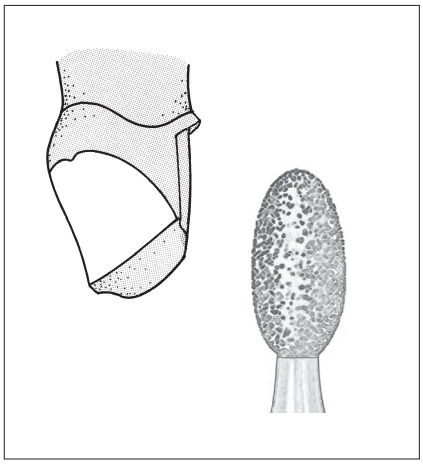


Fig 10-46 Lingual reduction: coarse-grit football-shaped diamond.



Fig 10-47 Lingual axial reduction: coarse-grit flat-end tapered diamond.



Fig 10-48 Axial wall and radial shoulder finishing: fine-grit flat-end tapered diamond.

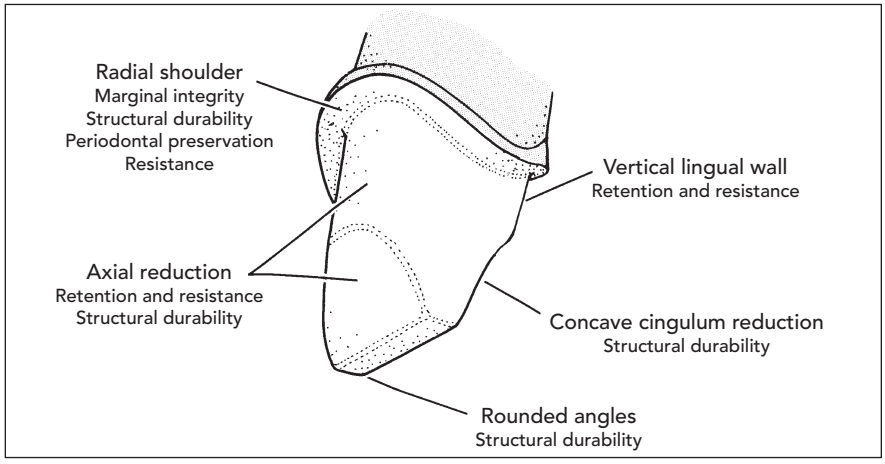


Fig 10-49 Features of an all-ceramic crown preparation and the function(s) served by each.

labial surface is reduced with the coarse-grit flat-end tapered diamond to a depth of 1.2 to 1.4 mm. This reduction extends around the labioproximal line angles and fades out on the lingual aspects of the proximal surfaces (Fig 10-45). The end of the coarse-grit flat-end tapered diamond bur will form the shoulder finish line, while the axial reduction is done with the sides of the diamond. The shoulder should be a minimum of 1.0 mm wide. Lingual reduction incisal to the cingulum is done with the coarse-grit football-shaped diamond, with care taken not to overreduce the junction between the cingulum and the lingual wall (apical to the cingulum) (Fig 10-46). Overshortening the lingual wall will reduce the retention of the preparation.

Reduction of the lingual axial surface is done with the coarse-grit flat-end tapered diamond (Fig 10-47). The wall should form a minimum taper with the gingival portion of the labial wall. The radial shoulder is at least 1.0 mm wide and should be a smooth continuation of the labial and proximal radial shoulders. All-ceramic crowns made over shoulder finish lines exhibit greater strength than those made over chamfers.[11,50] All of the axial walls should be smoothed with a fine-grit flat-end tapered diamond, accentuating the shoulder at the same time (Fig 10-48). All sharp angles should be rounded over at this time. The CP-11/12 modified binangle chisel is used to smooth the shoulder, removing any loose enamel rods at the cavosurface angle. Care must be taken not to create undercuts in the axial walls where they join the shoulder. The features of a preparation for an all-ceramic crown and the purpose served by each are shown in Fig 10-49.

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